Enantioselective Addition of Diethylzinc to Aldehydes Catalyzed by Optically Active C_2-Symmetrical Bis-β-amino Alcohols

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ABSTRACT The syntheses of new optically active C_2-symmetrical bis-β-amino alcohols **1-6** from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine are described. Especially attention is focused on bridges, which link the two β-amino alcohol units. These new chiral ligands have been successfully applied in the catalytic enantioselective addition of diethylzinc to aldehydes to give *sec*-alcohols in good yields with up to 95% enantiomeric excess. *Chirality 14:716–723, 2002.* © 2002 Wiley-Liss, Inc.

KEY WORDS: asymmetric reactions; diethylzinc; C_2-symmetrical bis-β-amino alcohols; aldehyde

The enantioselective addition reaction of organozinc reagents to aldehydes catalyzed by chiral ligands has received widespread attention because it is one of the most efficient methods for generating optically active secondary alcohols.[1–3] The importance of this structural feature arises mainly from the fact that it is part of many natural products or precursors as an important synthetic intermediate for various other functionalities, e.g., halide, amine, ester, and ether. For this purpose, a wide variety of chiral catalysts, i.e., amino alcohols,[4–10] diamines,[11–14] disulfonamides,[15–19] and diols[20–30] have been synthesized. Even though particularly efficient ligands for the enantiocontrolled ethylation of aldehydes by diethylzinc have been reported, it has been a continuous attracting subject for organic chemists to develop highly efficient ligands and meet practical application. Among the chiral ligands reported, β-amino alcohols are the most often-used chiral auxiliaries.[1–10] Since it was reported that the presence of the C_2 axis within a chiral auxiliary can serve the very important function of dramatically reducing the number of possible competing diastereomeric transition states,[31] many C_2-symmetrical auxiliaries have been synthesized and applied in the catalytic enantioselective addition reaction.[11–40] However, to our knowledge there are few C_2-symmetrical bis-β-amino alcohols being studied so far.[32–40] Recently, we reported that C_2-symmetrical bis-β-amino alcohol prepared from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine was an efficient chiral catalyst for the ethylation of aldehydes by diethylzinc.[41] In this article, we report the details of the syntheses of new optically active C_2-symmetrical bis-β-amino alcohols and their catalyzed asymmetric addition reaction of diethylzinc to aldehydes.

RESULTS AND DISCUSSION

Optically active C_2-symmetrical bis-β-amino alcohol **1** was obtained from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine[42] by coupling the two amine units via 1,2-dibromoethane/potassium carbonate in dry acetonitrile (Scheme 1).[39] Chiral C_2-symmetrical bis-β-amino alcohols **2** and **3** were prepared from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine by selective acetylation of the secondary amino group with acid chloride-Et_3N in benzene at room temperature (RT) to give the corresponding diamides, followed by reduction with excess $LiAlH_4$ in refluxing THF.[43]

Alternatively, optically active C_2-symmetrical bis-β-amino alcohols **4–6** can be conveniently produced from either (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine and the corresponding acid chloride as above or from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine and the corresponding dicarboxaldehyde followed by reduction with an excess of $LiAlH_4$ in refluxing THF.[44]

In order to compare the effect of asymmetric induction, "monomeric" β-amino alcohol analogs **7** and **8** were also synthesized in the same manner as that described in Scheme 1 (Scheme 2).

Excellent chiral inductions including asymmetric amplification has been reported for the asymmetric reaction of aldehydes with diethylzinc in the presence of chiral

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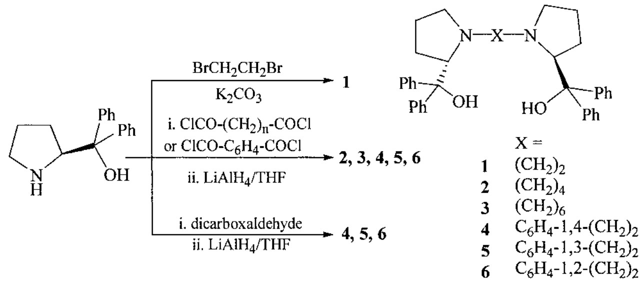
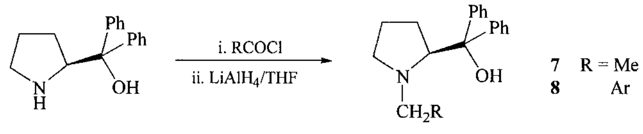


Scheme 1. Syntheses of chiral C_2-symmetrical bis-β-amino alcohols **1-6.**

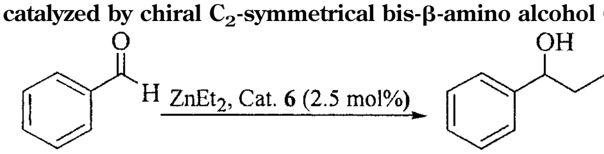
β-amino alcohol.[1–10,43,45–51] This reaction has been widely utilized for the examination of the efficiencies of chiral ligands.[1–10] In order to optimize the reaction conditions, we first examined the enantioselectivity of chiral ligand-catalyzed reactions of benzaldehyde with diethylzinc in the presence of C_2-symmetrical bis-β-amino alcohol **6** by varying solvent and temperature;[20] the results are summarized in Table 1. It was found that solvent drastically affected the e.e. (enantiomeric excess) of this asymmetric reaction. In contrast to the general results that hexane can give good e.e., a hexane solution of benzaldehyde reacting with 2.0 equiv. of diethylzinc in the presence of 2.5 mol% of chiral ligand **6** at 0°C can only give (S)-1-phenyl-1-propanol in 74% chemical yield with 10% e.e. (Table 1, Entry 1). However, when the same reaction was run in toluene under otherwise identical conditions, both the chemical yield and e.e. of the reaction product were considerably improved to 92% and 95%, respectively (Entry 2). The chemical yields and e.e.'s of the reactions decreased with an increase of hexane content when a mixed solvent of toluene/hexane was used (Entries 5, 6). This is because chiral ligand **6** can be dissolved in toluene easily and dissolved in hexane slightly. The same phenomena were also observed when benzene or benzene/hexane were used as solvent (Entries 7–9). For dichloromethane as the solvent, good yield and enantioselectivity were also obtained due to its excellent dissolvability (Entry 10). However, for THF and acetonitrile as solvent, because of their coordination to diethylzinc, slightly low enantioselectivities were observed (Entries 11, 12). Since many successfully enantiomeric syntheses were performed at low temperature,[20,21] we carried out this reaction at different temperatures. However, the results indicated that the enantioselectivities only increase slightly with the decrease of temperature (Entries 2–4). Therefore, we carried out the reaction in toluene at 0°C in the following experiment because it is easily available in the laboratory.

We next compared C_2-symmetrical bis-β-amino alcohol catalysts **1–6** with "monomeric" β-amino alcohol analogs **7**



Scheme 2. Syntheses of chiral β-amino alcohols **7** and **8.**

TABLE 1. Effect of solvent and temperature on the enantioselective addition of diethylzinc to benzaldehyde catalyzed by chiral C_2-symmetrical bis-β-amino alcohol 6[a]



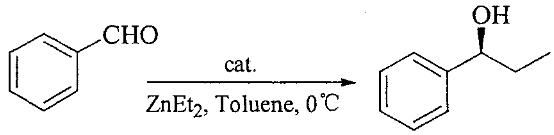
Entry	Solvent	Temperature (°C)	Yield (%)[b]	e.e. (%) (Config)[c]
1	Hexane	0	74	10 (S)
2	Toluene	0	92	95 (S)
3	Toluene	−20	92	95 (S)
4	Toluene	20	89	93 (S)
5	Toluene/Hexane (v/v, 1:1)	0	91	83 (S)
6	Toluene/Hexane (v/v, 1:3)	0	85	74 (S)
7	Benzene	0	83	85 (S)
8	Benzene/Hexane (v/v, 1:1)	0	82	72 (S)
9	Benzene/Hexane (v/v, 1:3)	0	78	67 (S)
10	Dichloromethane	0	83	93 (S)
11	THF	0	69	74 (S)
12	Acetonitrile	0	74	71 (S)

[a]The reactions were carried out at indicated temperature with chiral ligand **6**:benzaldehyde: diethylzinc = 0.125:5.0:10.0 (mmol).
[b]Isolated yield.
[c]The e.e. values were determined by GC and the configurations were determined by comparing the sign of specific rotation with the known compound.[49]

and **8** in the same enantioselective addition reaction by means of the above optimized conditions; the results are summarized in Table 2. For comparison purposes, the reaction was also carried out with (S)-(+)-1-methyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine (DPMPM) as chiral ligand.[42] (*S*)-1-Phenyl-1-propanol was preferentially obtained in 87–93% yields with e.e.'s varying from 3–95%. Interestingly, for alkylene bridge-linked C_2-symmetrical bis-β-amino alcohols **1–3**, the enantioselectivities decreased when compared with "monomeric" β-amino alcohol analog **7** (Entries 1–3 vs. 7). On the other hand, for xylylene bridge-linked C_2-symmetrical bis-β-amino alcohols **4–6,** enantioselectivities increased for ligand **6** and decreased for ligands **4** and **5** when compared with "monomeric" β-amino alcohol analog **8** (Entries 4–6 vs. 8). It was expected that chiral ligands **3** and **4,** which have longer bridges, would give high enantioselectivities, because the two β-amino alcohol groups have less repulsive interaction and act as two "monomeric" β-amino alcohols independently. Practically, it is chiral ligand **6**, which has a much more rigid and crowded structure, which gives the best enantioselectivity. Thus, a sterically more demanding structure in the C_2-symmetrical bis-β-amino alcohol **6** seems to be crucial for enhancing the enantioselectivity for catalytic asymmetric addition of diethylzinc to aldehydes[20,21] and the presence of a C_2 axis within a chiral auxiliary cannot necessarily induce high enantioselectivity.

During the investigation of the chiral amplification phe-

TABLE 2. Enantioselective addition of diethylzinc to benzaldehyde promoted by chiral ligands 1–8 and DPMPM[a]



Entry	Catalyst (mol%)	Yield (%)[b]	e.e (%) (Config.)[c]
1	**1** (2.5)	87	65 (S)
2	**2** (2.5)	92	58 (S)
3	**3** (2.5)	89	60 (S)
4	**4** (2.5)	87	28 (S)
5	**5** (2.5)	88	3 (S)
6	**6** (2.5)	92	95 (S)
7	**7** (5.0)	88	87 (S)
8	**8** (5.0)	89	40 (S)
9	DPMPM (5.0)	90	93 (S)

[a]The reactions were carried out in toluene at 0°C with benzaldehyde:diethylzinc = 5.0:10.0 (mmol).
[b]Isolated yield.
[c]The e.e. values were determined by GC and the configurations were determined by comparing the sign of specific rotation with the known compound.[49]

nomena in the chiral β-amino alcohol (2S)-3-*exo*-(dimethylamino)-isobornenol (DAIB) promoted alkylation of aldehydes with dialkylzinc, Noyori and colleagues[45,46] demonstrated the existence of equilibrium between monomeric and dimeric Zn-ligand complexes and it is the monomeric alkylzinc aminoalkoxide that acts as an active species to react with aldehydes. Molecular orbital calculations revealed that the asymmetric bias is generated in the intramolecular alkyl-transfer step in the "*anti*" or "*syn*" μ–O transition structure by differences among their stabilities. Since the *anti*-TS is much more stable than the *syn*-TS according to Noyori's mechanism,[47] the *anti*-configured 5/4/4 tricyclic transition states (*anti*-TS) is predominately produced. Based on their results, a possible mechanism with a 5/4/4-fused tricyclic transition state (*anti*-TS(S) and *anti*-TS(R)) for the ethylation of aldehyde is established to explain the stereochemical course and enantioselectivity for chiral C_2-symmetrical bis-β-amino-alcohols (Fig. 1). The attack of diethylzinc to benzaldehyde from the *Si*-face produces *anti*-TS(S) and from the *Re*-face produces *anti*-TS(R). Because of the 1,3-repulsive between Ph and Zn-linked Et in *anti*-TS(R), the bulkier phenyl group would take the energetically favorable equatorial position to form *anti*-TS(S). Thus, the transfer of the ethyl group occurs predominantly from the *Si*-face of PhCHO to furnish the observed (S)-1-phenyl-1-propanol.

The substituent on the nitrogen atom of the chiral catalysts may also affect the formation and stability of the transition state. In the case of the alkylene bridge-linked ligands **1–3,** because the flexibilities of the alkylene bridge may adjust Ph and bridge X in *anti*-TS(S) to minimize the steric repulsion between them, there is little difference in the e.e.'s of the products with the increase of bridge length. On the other hand, when they are compared with the "monomeric" β-amino alcohol analog **7,** the enantioselectivities decrease because the large bridge X inhibit more or less the formation of *anti*-TS(S).[51] For xylylene bridge-linked C_2-symmetrical bis-β-amino alcohols **4–6,** the enantioselectivities are much different because of their rigid bridges. For chiral ligands **4, 5,** the enantioselectivies decrease when they are compared with the "monomeric" β-amino alcohol analog **8** for the similar reason that the two β-amino alcohol units with a rigid bridge are far away from each other. However, for ligand **6** with a C_6H_4-1,2-$(CH_2)_2$ bridge, the enantioselectivity increases considerably when compared with ligand **8.** This may be due to the fact that the two β-amino alcohol units are close to each other in space, which enforce the repulsion among the two β-amino alcohols, Ph, and bridge X in *anti*-TS(R) and the *Re*-face of benzaldehyde is much more effectively blocked. In order to eliminate the steric effect, the ethyl group attacks benzaldehyde mostly from the *Si*-face to form *anti*-TS(S). Such improved enantioselectivity is observed. This explanation is reasonable when the monomeric amino-alcohols **7** and **8** were compared with DPMPM (Entries 7–9). For these ligands, the bulkiness of the N-substitutes are in the order **8** > **7** > DPMPM and the enantioselectivities decreased with DPMPM > **7** > **8.**[50]

The enantioselective addition reaction of diethylzinc to various aldehydes catalyzed by 2.5 mol% of chiral ligand **6** are summarized in Table 3. The results revealed that the addition reaction produced the corresponding *sec*-alcohols in high yields and 27–95% e.e. with S-configuration. When substituted aromatic aldehydes and naphthaldehyde were used as substrates, high e.e. values of the corresponding *sec*-alcohols were obtained in good yield (Entries 1–11) except for 4-pyridinecarboxaldehyde, which afforded the poorest e.e. (46%) (Entry 12). This can be interpreted by the competitive side catalytic cycle involving the basic heteroatom, which contributes to some minor catalytic pathways.[43] On the other hand, the reaction of *trans*-cinnamaldehyde, dodecylaldehyde, and cyclohexanecarboxaldehyde produced moderate e.e. values of the corresponding products (63, 60, and 65% e.e., respectively) (Entries 13–15). For nonylaldehyde as the substrate, low e.e. (27%) of 3-undecanol was obtained (Entry 16).

In conclusion, we have successfully synthesized six C_2-symmetric bis-β-amino alcohols from (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine. Enantioselective ethylation of aldehydes with diethylzinc was investigated in the presence of a catalytic amount of the chiral ligands and the effects of their structures on the enantioselectivities were

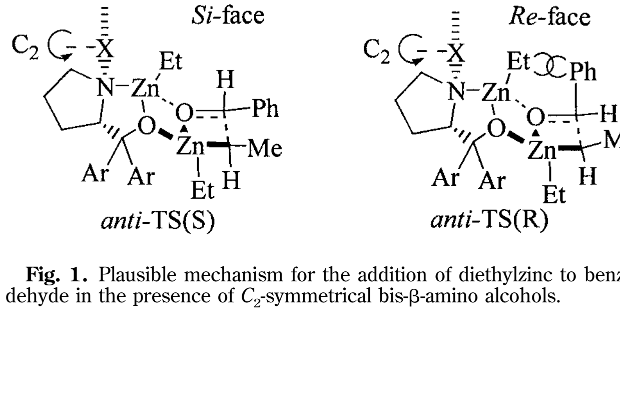
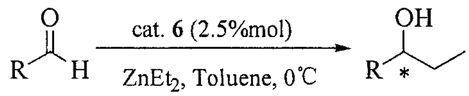


Fig. 1. Plausible mechanism for the addition of diethylzinc to benzaldehyde in the presence of C_2-symmetrical bis-β-amino alcohols.

TABLE 3. Enantioselective addition of diethylzinc to aldehydes catalyzed by chiral C_2-symmetrical bis-β-amino alcohol 6[a]



Entry	Substrate	Yield (%)[b]	e.e. (%) (Config.)[c]
1	Benzaldehyde	92	95 (S)
2	*o*-Anisaldehyde	89	90 (S)
3	*p*-Anisaldehyde	79	84 (S)
4	*o*-Chlorobenzaldehyde	90	85 (S)
5	*p*-Chlorobenzaldehyde	81	82 (S)
6	*p*-Tolualdehyde	86	80 (S)
7	*p*-Morpholinobenzaldehyde	91	90[d,e]
8	4-(Dimethylamino) benzaldehyde	89	89 (S)[d]
9	3,4-Dimethoxybenzaldehyde	78	83 (S)[d]
10	1-Naphthaldehyde	81	85 (S)[d]
11	2-Naphthaldehyde	83	87 (S)[d]
12	4-Pyridinecarboxaldehyde	78	46 (S)[d]
13	*Trans*-Cinnamaldehyde	79	63 (S)[d]
14	Dodecylaldehyde	83	60 (S)[f]
15	Cyclohexanecarboxaldehyde	86	65 (S)[f]
16	Nonylaldehyde	78	27 (S)[f]

[a]The reactions were carried out in toluene at 0°C with ligand **6**:aldehyde:diethylzinc = 0.125:5.0:10.0 (mmol).
[b]Isolated yield.
[c]Unless otherwise specified, the e.e. values were determined by GC and the configurations were determined by comparising the sign of specific rotation with known compounds.
[d]Determined by HPLC.
[e]The configuration was not determined.
[f]Determined by GC after acetylation.

examined. Chiral ligand **6** is the most efficient catalyst in this series. Its high enantioselectivity induction may rely on the steric congestion between the two β-amino alcohol units. This chiral ligand also promoted the asymmetric ethylation of a number of substituted benzaldehydes, naphthaldehydes, and aliphatic aldehydes. Enantiomeric excess of up to 95% was observed with good yield. Further applications of these ligands in other catalytic asymmetric reactions are in progress.

EXPERIMENTAL

All reactions were carried out under argon atmosphere. Melting points were taken on a Kofler melting apparatus and are uncorrected. ^1H and ^{13}C NMR spectra were measured with a Varian 200 or Bruker AM-400 NMR spectrometer with TMS as an internal reference (chemical shifts are expressed as values in ppm and coupling constants in hertz). Positive ion FAB mass spectra as 3-nitrobenzyl alcohol matrix were recorded on VG ZAB-HS mass spectrometer. Elemental analyses were performed on a Carlo-Erba-1106 elemental analyzer. Optical rotations were measured on a JASCO J-20C automatic polarimeter. Enantiomeric excess (e.e.) determination was carried out on a Varian CP-3380 GC instrument with FID as the detector, nitrogen as the carrier gas, and a Chrompack CP-Chirasil-DEX CB capillary column, or on a Varian SD-200 HPLC instrument with UV as the detector, hexane/2-propanol as eluent, and a Chiralcel-OD column. $ZnEt_2$ (1.0 M in hexanes) was purchased from Fluka (Buchs, Switzerland) and used as received. All solvents were dried by standard methods and aldehydes were purified by standard, published methods before use.

Synthesis of N,N′-1,2-ethane-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 1

To a solution of (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine (0.5 g, 1.98 mmol) and anhydrous potassium carbonate (1.1 g, 7.97 mmol) in dry acetonitrile (8 ml), 1,2-dibromoethane (0.185 g, 0.98 mmol, in 7 ml acetonitrile) was added dropwise within 30 min at 80°C. Stirring was continued at 80°C for 48 h under Ar atmosphere. The solvent was removed in vacuo. The residue was dissolved in dichloromethane (20 ml) and washed with water (2 × 20 ml). After drying over anhydrous sodium sulfate, filtrating, and evaporating under reduced pressure, the resulting crude solid was purified by flash chromatography (petroleum ether: ethyl acetate 5:1) to give 0.24 g white solid (46% yield). Mp. 174–175°C; $[\alpha]_D^{20}$ = -5.6 (c 1.05, CH_2Cl_2); ^1H NMR ($CDCl_3$, 200 MHz): δ ppm 1.38-1.62 (m, 6H), 1.65-1.70 (m, 2H), 1.72-1.97 (m, 4H), 2.16-2.22 (m, 2H), 2.32-2.39 (m, 2H), 3.69-3.75 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.75-4.90 (br, 2H), 7.08-7.32 (m, 12H), 7.48-7.53 (m, 4H), 7.64-7.68 (m, 4H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.12, 29.57, 53.64, 54.76, 68.86, 77.32, 125.36, 125.66, 126.00, 126.14, 127.94, 128.09, 146.98, 148.52; Positive ion FAB mass spectra *m/z*: 533 (M^++H); Anal. Calcd. for $C_{36}H_{40}N_2O_2$: C 81.17, H 7.57, N 5.26; Found: C 81.09, H 7.65, N 5.33.

Syntheses of Diamides of (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine: General Procedure

To a solution of acid chloride (1.18 mmol) and triethylamine (6.02 mmol) in anhydrous benzene (2 ml) was added (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine (0.61 g, 2.41 mmol) in anhydrous benzene (3 ml) at −10°C. The resulting mixture was stirred at RT for 4 h under Ar atmosphere. The precipitate was filtered off through Celite with rinsing by Et_2O (3 × 50 ml). The combined organic layers were washed with brine, saturated sodium hydrogen carbonate, and brine consecutively. After drying over anhydrous sodium sulfate, filtrating, and concentrating under reduced pressure, the crude product was purified by flash chromatography (petroleum ether : ethyl acetate 1:2) to give diamides of (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine.

Synthesis of N,N′-succinyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine)

Prepared from succinyl chloride in 89% yield; pale yellow solid; Mp. 96–98°C, $[\alpha]_D^{20}$ = +44.4 (c 1.00, CH_2Cl_2); ^1H NMR ($CDCl_3$, 200 MHz): δ ppm 0.72-0.95 (m, 2H), 1.45-1.68 (m, 2H), 1.91-2.17 (m, 4H), 2.34-2.40 (m, 2H), 2.79-2.85 (m, 2H), 2.89-2.97 (m, 2H), 3.48-3.53 (m, 2H), 5.17-5.23 (dd, *J* = 9.1, 4.9 Hz, 2H), 6.81 (br, 2H), 7.26-7.41 (m, 20H); Positive ion FAB mass spectra *m/z*: 589 (M^++H); Anal. Calcd. for $C_{38}H_{40}N_2O_4$: C 77.51, H 6.85, N 4.76; Found: C 77.63, H 6.92, N 4.59.

Synthesis of N,N′-adipoyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine)

Prepared from adipoyl chloride in 83% yield; pale yellow solid; Mp. 112–113°C; $[\alpha]_D^{20}$ = +31.2 (c 1.00, CH_2Cl_2); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 0.83-1.03 (m, 2H), 1.42-1.66 (m, 6H), 1.94-2.16 (m, 4H), 2.23-2.32 (m, 4H), 2.87-2.99 (m, 2H), 3.31-3.39 (m, 2H), 5.13-5.20 (dd, *J* = 9.4, 4.8 Hz, 2H), 7.00-7.43 (m, 20H); MS (EI)*m/z*: 434 (12), 364 (74), 346 (25), 236 (100), 183 (14), 105 (34), 76 (61); Positive ion FAB mass spectra *m/z*: 617 (M^++H); Anal. Calcd. for $C_{40}H_{44}N_2O_4$: C 77.88, H 7.19, N 4.54; Found: C 77.72, H 7.10, N 4.67.

Synthesis of N,N′-phthaloyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine)

Prepared from phthaloyl dichloride in 76% yield; pale yellow solid; Mp. 114–116°C; $[\alpha]_D^{20}$ = +13.6 (c 1.00, CH_2Cl_2); ^{1}H NMR($CDCl_3$, 200 MHz): δ ppm 1.26-1.42 (m, 2H), 1.61-1.75 (m, 2H), 1.88-2.05 (m, 2H), 2.14-2.27 (m, 2H), 2.66-2.79 (m, 2H), 3.23-3.32 (m, 2H), 6.68-6.73 (dd, *J* = 9.4, 4.9 Hz, 2H), 6.95 (br, 2H,), 7.24-7.64 (m, 24H); Positive ion FAB mass spectra *m/z*: 637 (M^++H); Anal. Calcd. for $C_{42}H_{40}N_2O_4$: C 79.22, H 6.33, N 4.40; Found: C 79.30, H 6.23, N 4.55.

Syntheses of C_2-Symmetrical Bis-β-amino Alcohols 2, 3, and 6 by $LiAlH_4$ Reduction of Diamides: General Procedure

To a stirred solution of the above-prepared diamides (1.02 mmol) in anhydrous THF (20 ml) at 0°C under Ar atmosphere, was added lithium aluminum hydride (4.20 mmol) in portionwise. The resulting suspension was heated under reflux for 2 h. When it was cooled to 0°C, water was added to quench the reaction. The white solid was removed by filtration with rinsing by ethyl acetate (2 × 20 ml). The filtrate was extracted with ethyl acetate (3 × 30 ml). The combined extracts were washed with brine, dried over anhydrous sodium sulfate, filtered, and evaporated under reduced pressure. The crude product was purified by flash chromatography to give a white solid.

Synthesis of N,N′-1, 4-Butane-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 2

Prepared from N,N′-succinyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) in 89% yield; white solid; Mp 163–164°C; $[\alpha]_D^{20}$ = +7.0 (c 1.00, CH_2Cl_2); ^{1}H NMR($CDCl_3$, 200 MHz): δ ppm 0.61-0.98 (m, 4H,), 1.55-1.70 (m, 8H), 1.80-1.92 (m, 4H), 2.24-2.32 (m, 2H), 3.05-3.14 (m, 2H), 3.68-3.74 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.6-5.0 (br, 2H), 7.11-7.31 (m, 12H), 7.50-7.59 (m, 8H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.39, 26.11, 29.52, 55.04, 55.96, 71.00, 77.65, 125.62, 125.66, 126.05, 126.14, 127.83, 127.98, 146.65, 148.27; Positive ion FAB mass spectra *m/z*: 561 (M^++H); Anal. Calcd. for $C_{38}H_{44}N_2O_2$: C 81.38, H 7.91, N 5.00. Found: C 81.45, H 7.87, N 4.93.

Synthesis of N,N′-1,6-Hexane-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 3

Prepared from N,N′-adipoyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine)in 81% yield; white solid; Mp. 127–128°C; $[\alpha]_D^{20}$ = +12.6 (c 1.00, CH_2Cl_2); ^{1}H NMR($CDCl_3$, 200 MHz): δ ppm 0.59-0.75 (m, 4H), 1.02-1.15 (m, 4H), 1.61-1.68 (m, 6H), 1.77-1.95 (m, 6H), 2.28-2.36 (m, 2H), 3.16-3.21 (m, 2H), 3.73-3.79 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.94 (br, 2H), 7.06-7.31 (m, 12H), 7.51-7.61 (m, 8H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.35, 26.68, 28.59, 29.59, 55.15, 56.10, 71.12, 77.59, 125.61, 125.63, 126.08, 126.11, 127.68, 127.97, 146.78, 148.32; Positive ion FAB mass spectra *m/z*: 589 (M^++H); Anal. Calcd. for $C_{40}H_{48}N_2O_2$: C 81.58, H 8.22, N 4.76; Found: C 81.50, H 8.27, N 4.82.

Synthesis of N,N′-α,α′-o-xylylene-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 6

Prepared from N,N′-Phthaloyl-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) in 86% yield; white solid; Mp. 110–112°C; $[\alpha]_D^{20}$ = +4.5 (c 0.53, CH_2Cl_2); ^{1}H NMR($CDCl_3$, 200 MHz): δ ppm 1.60-1.68 (m, 4H), 1.70-1.80 (m, 2H), 1.92-2.06 (m, 2H), 2.18-2.30 (m, 2H), 2.73-2.80 (m, 2H), 2.95, 3.08 (AB system, *J* = 12.6 Hz, 4H), 3.96-4.00 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.80 (br, 2H), 7.05-7.37 (m, 16H), 7.54-7.71 (m, 8H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.25, 29.60, 55.44, 57.07, 71.20, 77.96, 125.44, 125.61, 126.22, 126.29, 126.49, 128.02, 128.10, 128.76, 137.23, 146.73, 147.96; Positive ion FAB mass spectra *m/z*: 609 (M^++H); Anal. Calcd. for $C_{42}H_{44}N_2O_2$: C 82.85, H 7.29, N 4.60; Found: C 82.80, H 7.26, N 4.67.

Syntheses of C_2-Symmetrical Bis-β-amino Alcohols 4–6 From the Corresponding Dicarboxaldehyde: General Procedure

A mixture solution of dicarboxaldehyde (0.20 g, 1.49 mmol), (S)-2-(1-hydroxy-1,1- diphenylmethyl)-pyrrolidine (0.78 g, 3.08 mmol) and molecular sieves (0.7 g) in anhydrous CH_2Cl_2 (10 ml) was stirred at RT under Ar atmosphere until the reaction was finished (petroleum ether : ethyl acetate 3:1). The mixture was filtered off through a pad of celite and the solvent evaporated on a Rotavapor. The residue was dissolved in anhydrous THF (20 ml) and to this solution was added lithium aluminum hydride (140 mg, 3.6 mmol) portionwise and refluxed for 4 h under Ar atmosphere. The solution was cooled to RT and sequentially treated with water. The white solid was removed by filtration with rinsing by ethyl acetate (2 × 20 ml). The filtrate was extracted with CH_2Cl_2 (3 × 30 ml) and the combined organic layers were dried over anhydrous sodium sulfate, filtered, and evaporated under reduced pressure. The crude product was purified by flash chromatography to give C_2-symmetrical bis-β-amino alcohols.

Synthesis of N,N′-α,α′-p-xylylene-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 4

Prepared from terephthaldicarboxaldehyde and (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine in 71% yield; white solid; Mp 226–227°C; $[\alpha]_D^{20}$ = +25.6 (c 0.50, CH_2Cl_2); ^{1}H NMR($CDCl_3$, 200 MHz): δ ppm 1.60-1.68 (m, 4H), 1.74-1.81 (m, 2H), 1.93-2.01 (m, 2H), 2.31-2.37 (m, 2H), 2.87-2.91 (m, 2H), 2.98, 3.17 (AB system, *J* = 12.5 Hz, 4H), 3.96-3.99 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.95 (br, 2H), 6.93 (s, 4H), 7.11-7.21 (m, 4H), 7.28-7.34 (m, 8H), 7.60-7.74 (m, 8H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.09, 29.77, 55.49, 60.23, 70.57, 77.68, 125.53, 125.59, 126.20, 126.35, 128.04, 128.15, 128.38, 138.24, 146.66, 148.03; Positive ion FAB mass spectra *m/z*: 609 (M^++H); Anal. Calcd. for $C_{42}H_{44}N_2O_2$: C 82.85, H 7.29, N 4.60; Found: C 82.93, H 7.22, N 4.65.

Synthesis of N,N′-α,α′-m-xylylene-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 5

Prepared from isophthalaldehyde and (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine in 75% yield; white solid; Mp 170–171°C; $[\alpha]_D^{20}$ = +18.8 (c 0.50, CH_2Cl_2); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 1.58-1.68 (m, 4H), 1.73-1.80 (m, 2H), 1.92-2.00 (m, 2H), 2.27-2.40 (m, 2H), 2.83-2.88 (m, 2H), 2.95, 3.16 (AB system, *J* = 12.6 Hz, 4H), 3.95-4.02 (dd, *J* = 9.4, 4.8 Hz, 2H), 4.95 (br, 2H), 7.05-7.37 (m, 16H), 7.54-7.71 (m, 8H); ^{13}C NMR ($CDCl_3$, 100 MHz): δ ppm 24.18, 29.61, 55.47, 60.42, 70.59, 77.69, 125.57, 125.64, 126.22, 126.36, 127.09, 127.63, 128.06, 128.16, 128.70, 139.51, 146.63, 148.09; Positive ion FAB mass spectra *m/z*: 609 (M^++H); Anal. Calcd. for $C_{42}H_{44}N_2O_2$: C 82.85, H 7.29, N 4.60; Found: C 82.93, H 7.35, N 4.53.

Synthesis of N,N′-α,α′-o-xylylene-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) 6

Prepared from phthalic dicarboxaldehyde and (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine to give N,N′-α,α′-*o*-xylylene-bis-((S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine) **6** which has spectra the same as that described above for **6**.

Syntheses of (S)-1-Acyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine and (S)-1-Benzoyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine: General Procedure

To a solution of acyl chloride (0.13 ml, 1.18 mmol) and triethylamine (6.02 mmol) in anhydrous benzene (2 ml) was added (S)-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine (0.61 g, 2.41 mmol) in anhydrous benzene (3 ml) at −10°C. The resulting mixture was stirred at RT for 4 h under Ar atmosphere. The precipitate was filtered through Celite with rinsing by Et_2O (3 × 50 ml). The combined organic layers were washed with brine, saturated sodium hydrogen carbonate, and brine consecutively. After drying over anhydrous sodium sulfate, filtrating, and concentrating under reduced pressure, the crude product was purified by flash chromatography to give (S)-1-acyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine.

Synthesis of (S)-1-Acyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine

Prepared from acyl chloride in 87% yield; pale yellow solid; Mp. 95–96°C; $[\alpha]_D^{20}$ = -35.4 (c 1.00, CH_2Cl_2); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 0.78-0.92 (m, 1H), 1.51-1.57 (m, 1H), 1.93-2.19 (m, 2H), 2.08 (s, 3H), 2.85-2.97 (m, 1H), 3.27-3.40 (m, 1H), 5.17-5.23 (dd, *J* = 9.4, 4.7 Hz, 1H), 7.27-7.42 (m, 10H); Positive ion FAB mass spectra *m/z*: 296 (M^++H); Anal. Calcd. for $C_{19}H_{21}NO_2$: C 77.25, H 7.17, N 4.74; Found: C 77.43, H 7.20, N 4.92.

Synthesis of (S)-1-Benzoyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine

Prepared from benzoyl chloride in 83% yield; pale yellow solid; Mp. 179–180°C; $[\alpha]_D^{20}$ = -17.1 (c 1.00, CH_2Cl_2); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 1.43-1.62 (m, 2H), 1.87-2.22 (m, 2H), 2.78-2.95 (m, 1H), 3.20-3.34 (m, 1H), 5.32-5.41 (dd, *J* = 9.4, 4.8 Hz, 1H), 6.8-7.0 (br, 1H), 6.89-7.65 (m, 15H); Positive ion FAB mass spectra *m/z*: 358 (M^++H); Anal. Calcd. for $C_{24}H_{23}NO_2$: C 80.63, H 6.49, N 3.92; Found: C 80.69, H 6.70, N 3.87.

Syntheses of (S)-1-Ethyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine 7 and (S)-1-Benzyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine 8: General Procedure

To a stirred solution of (S)-1-acyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine (1.02 mmol) in anhydrous THF (20 ml) at 0°C under Ar atmosphere was added lithium aluminum hydride (2.10 mmol) portionwise. The resulting suspension was heated under reflux for 2 h. When it was cooled to 0°C, water was added to quench the reaction. The white solid was removed by filtration with rinsing by ethyl acetate (2 × 20 ml) and the filtrate was extracted with ethyl acetate (3 × 30 ml). The combined extracts were dried over sodium sulfate, filtered, and evaporated under reduced pressure. The crude product was purified by flash column chromatography (silica gel, petroleum ether : ethyl acetate) to give a white solid.

Synthesis of (S)-1-Ethyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine 7

Prepared from (S)-1-acyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine in 81% yield; white solid; Mp. 78–79°C; $[\alpha]_D^{20}$ = +6.3 (c 0.64, CH_2Cl_2); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 0.78 (t, *J* = 7.2 Hz, 3H), 1.62-1.76 (m, 3H), 1.84-2.08 (m, 3H), 2.33-2.45 (m, 1H), 3.17-3.27 (m, 1H), 3.76-3.80 (dd, *J* = 9.2, 4.8 Hz, 1H), 4.70-5.15 (br, 1H), 7.11-7.32 (m, 6H), 7.52-7.62 (m, 4H); Positive ion FAB mass spectra *m/z*: 282 (M^++H); Anal. Calcd. for $C_{19}H_{23}NO$: C 81.09, H 8.24, N 4.98; Found: C 80.93, H 8.18, N 5.09.

Synthesis of (S)-1-Benzyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine 8

Prepared from (S)-1-benzoyl-2-(1-hydroxy-1,1-diphenylmethyl)-pyrrolidine in 86% yield; white solid; Mp. 115–116°C; $[\alpha]_D^{20}$ = +67.0 (c 1.00, CH_2Cl_2) (lit.[42] Mp. 113–115°C, $[\alpha]_D^{20}$ = +76.2 (c 1.6, CH_2Cl_2)); ^{1}H NMR ($CDCl_3$, 200 MHz): δ ppm 1.60-1.82 (m, 3H), 1.94-2.05 (m, 1H), 2.31-2.44 (m, 1H), 2.91-2.96 (m, 1H), 3.07, 3.28 (AB system, *J* = 12.6 Hz, 2H), 3.96-4.03 (dd, *J* = 9.6, 4.8 Hz, 1H), 4.85-5.10 (br, 1H), 7.05-7.35 (m, 10H), 7.59-7.63 (m, 2H), 7.73-7.77 (m, 3H); Positive ion FAB mass spectra *m/z*: 344 (M^++H); Calcd. for $C_{24}H_{25}NO$: C 83.92, H 7.34, N 4.08; Found: C 84.03, H 7.28, N 4.21.

General Procedure for Asymmetric Addition of Diethylzinc to Benzaldehyde

To a solution of catalyst **6** (1.0 mmol) in toluene (4 ml) at 0°C was added a 1.0 M solution (10.0 ml, 10.0 mmol) of diethylzinc in hexane. After stirring for 30 min at 0°C, freshly distilled benzaldehyde (5.0 mmol) was added. The reaction mixture was stirred at 0°C for 12 h. After addition of 1N HCl (10 ml), the phases were separated. The water layer was extracted with ethyl acetate (3 × 20 ml). The combined organic layers were washed with brine, dried over anhydrous sodium sulfate, filtrated, and concentrated under reduced pressure. After purification by flash chromatography (petroleum ether : ethyl acetate 5:1), the e.e. was determined by gas chromatography.

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